Math 252 - Linear Algebra Review

Rutgers University, Summer 2022

This material will **not** be covered in class; it's part of the prerequisites for Math 252 and we don't have time to review it. But it will be heavily used in much of the course, particularly on Chapter 3, therefore students are urged to study it by themselves and make sure they are comfortable with the topic. Homeworks and quizzes will include topics from this material.

1 Matrix Basics

A matrix is a table of numbers (real or complex), which can be enclosed by parentheses or square brackets:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrices of the same size can be summed. This is done elementwise:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 0 & 3 \\ 4 & -2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 6 \\ 8 & 3 & 6 \end{bmatrix}$$

A matrix can be multiplied by a scalar (a real or complex number). This is done elementwise:

$$2\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Multiplication of two matrices is only defined if the first has as many columns as the second has rows. The resulting matrix has as many rows as the first and as many columns as the second, and each of its elements is a sum of products of elements of the corresponding row of the first and column of the second:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 0 & -2 \\ 3 & 0 \end{bmatrix} &= \begin{bmatrix} 1\cdot(-1)+2\cdot 0+3\cdot 3 & 1\cdot 4+2\cdot(-2)+3\cdot 0 \\ 4\cdot(-1)+5\cdot 0+6\cdot 3 & 4\cdot 4+5\cdot(-2)+6\cdot 0 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 0 \\ 14 & 6 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \begin{bmatrix} -1 & 4 \\ 0 & -2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} &= \begin{bmatrix} -1\cdot 1+4\cdot 4 & -1\cdot 2+4\cdot 5 & -1\cdot 3+4\cdot 6 \\ 0\cdot 1-2\cdot 4 & 0\cdot 2-2\cdot 5 & 0\cdot 3-2\cdot 6 \\ 3\cdot 1+0\cdot 4 & 3\cdot 2+0\cdot 5 & 3\cdot 3+0\cdot 6 \end{bmatrix} \\ &= \begin{bmatrix} 15 & 18 & 21 \\ -8 & -10 & -12 \\ 3 & 6 & 9 \end{bmatrix} \end{aligned}$$

A vector in n-dimensional space, that is, an element of $\mathbb{R}^n$ or $\mathbb{C}^n$, is usually written as an $n \times 1$ matrix (a column matrix). An $n \times n$ square matrix can then pre-multiply it and produce another $n \times 1$ vector. Such an operation is called a **linear transformation** of the vector:

$$\begin{bmatrix} * & * & \dots & * \\ * & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & * \end{bmatrix} \begin{bmatrix} * \\ * \\ \vdots \\ * \end{bmatrix} = \begin{bmatrix} * \\ * \\ \vdots \\ * \end{bmatrix}$$

The **identity matrix** in n dimensions is the $n \times n$ square matrix that only has nonzero entries in the main diagonal, where all entries are 1. It corresponds to the identity transformation of a vector, as can be easily checked by multiplying it with any vector:

$$I = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix} \quad ; \quad Iv = v \text{ for all } v \in \mathbb{C}^n$$

The **determinant** is a number associated to every square matrix. We can denote it by the word det in front of the matrix, or by vertical bars around it. For a 2×2 matrix, compute the product of the main diagonal and subtract that of the secondary diagonal:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

For a 3×3 matrix, compute the products of triples of elements pertaining each to one row and one column. There are 6 such triples. 3 of them get added and the other 3 subtracted:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - afh - bdi$$

We won't find the need to consider determinants of larger matrices.

A square matrix A with nonzero determinant admits an **inverse matrix** A^{-1}, that is, a matrix such that

$$AA^{-1} = A^{-1}A = I \qquad (\det A \neq 0)$$

To compute the inverse of a 2×2 matrix, switch the elements in the main diagonal, flip the signs in the secondary diagonal, and divide the entire matrix by the determinant of the original:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

To compute the inverse of a 3×3 matrix, you could apply the **cofactor construction** below, which uses 2×2 subdeterminants from the original matrix (notice the patterns and be careful with minus signs):

$$A^{-1} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^{-1} = \frac{1}{\det A} \begin{bmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} & -\begin{vmatrix} b & c \\ h & i \end{vmatrix} & \begin{vmatrix} b & c \\ e & f \end{vmatrix} \\ -\begin{vmatrix} d & f \\ g & i \end{vmatrix} & \begin{vmatrix} a & c \\ g & i \end{vmatrix} & -\begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ \begin{vmatrix} d & e \\ g & h \end{vmatrix} & -\begin{vmatrix} a & b \\ g & h \end{vmatrix} & \begin{vmatrix} a & b \\ d & e \end{vmatrix} \end{bmatrix}$$

Once finished with the computation, always check that your inverse satisfies $AA^{-1} = A^{-1}A = I$. Due to the complexity of this calculation, it will not show up on timed exams and quizzes, but may be needed on homeworks.

2 Linear Systems

A **linear system** of m equations in n unknowns $x_1, \dots, x_n$ is a system of the form

$$\begin{cases} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & = & b_2 \\ & & & & \vdots & & & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & = & b_m \end{cases}$$

for given scalars a_{ij} and b_i. It can be converted to matrix notation

$$Ax = b$$

if we consider the **coefficients matrix** A, the **nonhomogeneous term vector b** and the **unknowns vector x**:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad , \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad , \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Example. The system below is equivalent to the following matrix formulation:

$$\begin{cases} 2x + y &= 3 \\ x - 3y &= -5 \end{cases} \quad \iff \quad \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

Suppose $m = n$, that is, A is a square matrix, and we have as many unknowns as equations in the system. If $\det A \neq 0$, the system always has exactly one solution $\mathbf{x}$ for each nonhomogeneous term $\mathbf{b}$, because we can write down its solution using A^{-1} (even though this would be an impractical way of solving it):

$$A\mathbf{x} = \mathbf{b} \implies \mathbf{x} = A^{-1}\mathbf{b}$$

If $\det A = 0$, the system is called **singular**, and it may have either zero or infinitely many solutions, depending on the nonhomogeneous term $\mathbf{b}$.

Example (singular 2×2 system). A 2×2 matrix can only be singular if one of its rows is a multiple of the other (see next section). If the elements of $\mathbf{b}$ on the right-hand side of the system are related in a way consistent with this relation of multiplicity on the left-hand side, then our system consists of two equations which say the same thing. So it will have solutions, but there will be no way to find just one pair of solutions, only to extract one variable in terms of the other:

$$A = \begin{bmatrix} -3 & 1 \\ -6 & 2 \end{bmatrix}, \ \mathbf{b} = \begin{bmatrix} 4 \\ 8 \end{bmatrix} \implies \begin{cases} -3x + y &= 4 \\ -6x + 2y &= 8 \end{cases} \implies y = 3x + 4 \ \text{ (infinitely many solutions)}$$

But if the elements of $\mathbf{b}$ aren't compatible with the multiplicity relation on the left-hand side, there can't be any solution, since the two equations will be contradictory:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \ \mathbf{b} = \begin{bmatrix} 5 \\ 9 \end{bmatrix} \implies \begin{cases} x + 2y &= 5 \\ 3x + 6y &= 9 \end{cases} \implies x = 5 - 2y \ \text{ and } \ x = 3 - 2y \ \text{ (no solutions)}$$

When $\mathbf{b} = 0$ there is always a solution $\mathbf{x} = 0$ to the system $A\mathbf{x} = \mathbf{b}$. Thus, if it's a singular system, there will be infinitely many solutions. This observation will be pertinent in the context of finding eigenvectors: the system that finds them always has $\mathbf{b} = 0$.

3 Linear Dependence and Independence

We say that vectors $v_1, \dots, v_n \in \mathbb{C}^n$ satisfy a **linear relation** when there exists a nontrivial linear combination of them that results in 0, that is, there exist scalars $a_1, \dots, a_n \in \mathbb{C}^n$ *not all equal to zero* such that

$$a_1 v_1 + \dots + a_n v_n = 0 \tag{1}$$

In this case, we call the vectors **linearly dependent**. Otherwise, we call them **linearly independent**. If we label the elements of each of these vectors as such:

$$v_1 = \begin{bmatrix} c_{11} \\ \vdots \\ c_{1n} \end{bmatrix}, \ v_2 = \begin{bmatrix} c_{21} \\ \vdots \\ c_{2n} \end{bmatrix}, \ \cdots, \ v_n = \begin{bmatrix} c_{n1} \\ \vdots \\ c_{nn} \end{bmatrix}$$

then the linear relation (**??**) can be written as an $n \times n$ system:

$$\left\{\begin{array}{ccccccccc} c_{11}a_1 & + & c_{12}a_2 & + & \ldots & + & c_{1n}a_n & = & 0 \\ c_{21}a_1 & + & c_{22}a_2 & + & \ldots & + & c_{2n}a_n & = & 0 \\ & & & & \vdots & & & & \\ c_{n1}a_1 & + & c_{n2}a_2 & + & \ldots & + & c_{nn}a_n & = & 0 \end{array}\right.$$

Suppose we were trying to decide if the vectors $v_1, \ldots, v_n$ are linearly dependent or independent. We would write down this system and think of the scalars a_i as the unknowns. Since the vector term $\mathbf{b}$ for this system is null, it always has at least one solution $a_1 = \cdots = a_n = 0$, but in order for linear dependence to occur we must have a solution in which not all a_i's are null. This happens if and only if the matrix formed from the vectors $v_1, \ldots, v_n$ is singular, in which case there will actually be infinitely many solutions.

$$\det \begin{bmatrix} c_{11} & c_{12} & \ldots & c_{1n} \\ c_{21} & c_{22} & \ldots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \ldots & c_{nn} \end{bmatrix} = 0 \quad \Longleftrightarrow \quad v_1, \ldots, v_n \text{ are linearly dependent}$$

Example. The vectors

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \; v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \; v_3 = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

are linearly dependent. We could verify this by computing

$$\det \begin{bmatrix} 1 & 4 & 5 \\ 2 & 5 & 7 \\ 3 & 6 & 9 \end{bmatrix} = 0$$

or also by realizing that there is a nontrivial linear relation among them:

$$v_1 + v_2 - v_3 = 0$$

that is, the third is the sum of the other two.

Example. The vectors

$$v_1 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \; v_2 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \; v_3 = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$$

are linearly independent. We could verify this by computing

$$\det \begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 0 \\ 0 & 3 & 5 \end{bmatrix} = 2 \neq 0$$

or also by verifying that the null solution is the only solution to the system

$$\left\{\begin{array}{l} y + 2z = 0 \\ 2x = 0 \\ 3y + 5z = 0 \end{array}\right.$$

Indeed, the second equation already gives $x = 0$, and the first says $y = -2z$, which plugged into the third gives $-6z + 5z = 0 \;\Rightarrow\; z = 0 \;\Rightarrow\; y = 0$.

Note that we have only stated the notion of linear dependence and independence for the case when the number of vectors being considered is equal to their dimension. The definition generalizes for the case in which it's not. Then things can still be reduced to a linear system, but it will not have as many unknowns as equations, and thus we can't put the matter simply in terms of the coefficients matrix being singular (which requires it to be square). We'd need to discuss its **rank** in this context. See any linear algebra book for more on this.

4 Eigenstuff

Let A be an $n \times n$ square matrix with real or complex entries. An **eigenvector** of A corresponding to the **eigenvalue** $\lambda \in \mathbb{C}$ is a vector $v \in \mathbb{C}^n$, different from the zero vector, such that

$$Av = \lambda v$$

Not all numbers λ are eigenvalues, only those for which the equation above has a (nonzero) solution v. Since this equation can also be written as

$$(A - \lambda I)v = 0$$

which is an $n \times n$ linear system, we see that a necessary and sufficient condition for such a nonzero solution v to exist is that the determinant of the coefficients matrix of this system be zero:

$$\det(A - \lambda I) = 0$$

(otherwise it would be a nonsingular system, and the only solution would be $v = 0$). Now here's what the matrix $A - \lambda I$ looks like:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \qquad \Longrightarrow \qquad A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{bmatrix}$$

Computing its determinant produces an n-degree polynomial equation whose solutions are the eigenvalues of A. In general they can be complex numbers even when A is real, and there could be repeated eigenvalues (repeated roots of the polynomial equation).

Eigenvectors corresponding to different eigenvalues are always linearly independent. Eigenvectors corresponding to the same eigenvalue form a linear subspace called the **eigenspace**. That is, we can take their multiples or add them together and we get other eigenvectors. Because of that, when asked to find the eigenvectors corresponding to some eigenvalue, we can't list them all; we only need a **complete set** of them, which is to say just enough of them to **span** the entire eigenspace. If we know the dimension d of this eigenspace, we only need to find d linearly independent eigenvectors.

When an $n \times n$ matrix has n distinct eigenvalues, to each one we'll find only a 1-dimensional subspace of corresponding eigenvectors, that is, for each eigenvalue we only need one eigenvector. But when there are repeated eigenvalues, any one could have two or more independent eigenvectors. In this case the sum of the dimensions of each space of eigenvectors doesn't even need to be all of n; for example, in an extreme case, a matrix could have just one eigenvalue and only a 1-dimensional space of corresponding eigenvectors.

Finding eigenvectors: Substitute each eigenvalue λ found above into the matrix equation $\det(A - \lambda I)v = 0$, writing out all elements and treating the entries of v as unknowns. Expand out this equation into an $n \times n$ system and solve. Keep in mind that complex numbers will be involved here when the eigenvalue is complex. Since the coefficients matrix of this system has zero determinant, there must be infinitely many solutions, so at least one of the equations is superfluous. This means you can't expect to completely solve for all unknowns; instead, as you solve for each one in terms of the others, you'll find a set of linear relations satisfied by them, until you run out of equations. You are then free to choose the values of the remaining unknowns, which will yield many possible eigenvectors. The number of free variables left is the dimension of the eigenspace.

Example (2×2 matrix, two different real eigenvalues). Let's find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

We find the eigenvalues by solving

$$0 = \begin{vmatrix} 1 - \lambda & 2 \\ 4 & 3 - \lambda \end{vmatrix} = (1 - \lambda)(3 - \lambda) - 2 \cdot 4 = \lambda^2 - 4\lambda + 3 - 8 = \lambda^2 - 4\lambda - 5 \Rightarrow \lambda = 5 \text{ or } -1$$

Let's denote them by $\lambda_1 = 5$ and $\lambda_2 = -1$. We find corresponding eigenvectors by plugging in these λ's in the eigenvector equation:

$$\begin{bmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For λ_1, we get

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \left\{ \begin{array}{rcl} -4x+2y &=& 0 \\ 4x-2y &=& 0 \end{array} \right.$$

These two equations are multiples of each other, as they should be, since this system is supposed to be singular. Therefore we can work with any one of them, finding a relation between x and y:

$$y = 2x$$

but we won't be able to solve for x. We can choose any (nonzero) value for x, for example $x = 1$, and get a corresponding y from $y = 2x$. We obtain an eigenvector corresponding to $\lambda_1 = 5$:

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

We could have chosen many other values for x and gotten many other eigenvectors:

$$v_1 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad , \quad \begin{bmatrix} -1 \\ -2 \end{bmatrix} \quad , \quad \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \quad , \quad \ldots$$

This simply reflects the fact that the set of eigenvectors corresponding to an eigenvalue is a vector subspace. In general, choose values that simplify all components as much as possible.

Similarly for λ_2 we find

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \left\{ \begin{array}{rcl} 2x+2y &=& 0 \\ 4x+4y &=& 0 \end{array} \right. \implies y = -x$$

and we can take

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

It's a good idea to check that the eigenvector equations $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ are satisfied:

$$Av_1 = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 2 \cdot 2 \\ 4 \cdot 1 + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \lambda_1 v_1 \checkmark$$

$$Av_2 = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 2 \cdot (-1) \\ 4 \cdot 1 + 3 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = (-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \lambda_2 v_2 \checkmark$$

Example (2×2 matrix, two complex eigenvalues). Let's find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix}$$

Finding the eigenvalues (use the quadratic formula):

$$0 = \begin{vmatrix} 3-\lambda & -2 \\ 4 & -1-\lambda \end{vmatrix} = (3-\lambda)(-1-\lambda) - (-2)\cdot 4 = \lambda^2 - 2\lambda - 3 + 8 = \lambda^2 - 2\lambda + 5$$

$$\Rightarrow \lambda = \frac{1}{2}(2 \pm \sqrt{4 - 4 \cdot 1 \cdot 5}) = \frac{1}{2}(2 \pm 4i) = 1 \pm 2i$$

We have $\lambda_1 = 1 + 2i$, $\lambda_2 = 1 - 2i$. To find the eigenvectors, plug them into

$$\begin{bmatrix} 3-\lambda & -2 \\ 4 & -1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For λ_1,

$$\begin{bmatrix} 2-2i & -2 \\ 4 & -2-2i \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} (2-2i)x - 2y &= 0 \\ 4x + (-2-2i)y &= 0 \end{cases}$$

As always for 2×2 systems, we can use either equation to solve for one unknown in terms of the other. They are equivalent, even though they might not look like it when complex numbers are involved. Use the first:

$$y = (1-i)x$$

If we choose $x = 1$, we get $y = 1 - i$, thus

$$v_1 = \begin{bmatrix} 1 \\ 1-i \end{bmatrix}$$

For λ_2, we could use the same approach, but there is a shortcut we can take. For matrices with real entries, it's always the case that eigenvectors corresponding to *conjugate* eigenvalues are also *conjugate* (*conjugation* means flipping the sign of i). This is because we can take the complex conjugate of the relation:

$$Av_1 = \lambda_1 v_1 \implies A\overline{v_1} = \overline{\lambda_1 v_1} = \lambda_2 \overline{v_1} \implies v_2 = \overline{v_1}$$

So we immediately know that we can take

$$v_2 = \begin{bmatrix} 1 \\ 1+i \end{bmatrix}$$

Checking the eigenvalue relations in this case is a good exercise with multiplication of complex numbers:

$$Av_1 = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1-i \end{bmatrix} = \begin{bmatrix} 3 \cdot 1 - 2 \cdot (1-i) \\ 4 \cdot 1 - 1 \cdot (1-i) \end{bmatrix} = \begin{bmatrix} 1+2i \\ 3+i \end{bmatrix}$$

$$\lambda_1 v_1 = (1+2i) \begin{bmatrix} 1 \\ 1-i \end{bmatrix} = \begin{bmatrix} 1+2i \\ (1+2i)(1-i) \end{bmatrix} = \begin{bmatrix} 1+2i \\ 1-i+2i-2i^2 \end{bmatrix} = \begin{bmatrix} 1+2i \\ 1+i+2 \end{bmatrix} = \begin{bmatrix} 1+2i \\ 3+i \end{bmatrix} \checkmark$$

We don't need to check that $Av_2 = \lambda_2 v_2$ since it follows by conjugation.

Example (2×2 matrix, two repeated eigenvalues, 1-dim eigenspace). Consider

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$$

Finding the eigenvalues:

$$0 = \begin{vmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{vmatrix} = (2-\lambda)(4-\lambda) - 1 \cdot (-1) = \lambda^2 - 6\lambda + 8 + 1 = \lambda^2 - 6\lambda + 9 = (\lambda-3)^2 \Rightarrow \lambda = 3$$

There is only one root, even though this equation is quadratic (we call that a *repeated root*). We may now find just one or two independent eigenvectors corresponding to this eigenvalue. It will depend on how many degrees of freedom are left after we extract full information from the system that determines eigenvectors:

$$\begin{bmatrix} 2-3 & 1 \\ -1 & 4-3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} -x+y &= 0 \\ -x+y &= 0 \end{cases}$$

We find one relation between x and y. We can take one of them to be arbitrary, and the other is determined by the equation $x = y$. This means the space of eigenvectors corresponding to $\lambda = 3$ is 1-dimensional, spanned for example by the vector

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Example (2×2 matrix, two repeated eigenvalues, 2-dim eigenspace). Consider

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

Finding the eigenvalues:

$$0 = \begin{vmatrix} 5-\lambda & 0 \\ 0 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 \ \Rightarrow \ \lambda = 5$$

There is only one root. Again we may find just one or two independent eigenvectors corresponding to it:

$$\begin{bmatrix} 5-5 & 0 \\ 0 & 5-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Longrightarrow \begin{cases} 0 = 0 \\ 0 = 0 \end{cases}$$

We find a trivial system! Any values of x and y will satisfy these two equations, so we have 2 degrees of freedom in picking our eigenvectors. In other words, the space of eigenvectors corresponding to $\lambda = 5$ is 2-dimensional (and must be the whole plane). As for two representative linearly independent eigenvectors, we can choose

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ , \ v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

It's no surprise that every vector is an eigenvector of this matrix, since the matrix is simply a multiple of the identity, so when we apply it to any vector of course we obtain a multiple of it:

$$Av = 5Iv = 5v \quad , \quad v \in \mathbb{R}^2$$

Example (3×3 matrix, one real and two complex eigenvalues). Consider

$$A = \begin{bmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ -3 & 0 & 0 \end{bmatrix}$$

Finding the eigenvalues:

$$\begin{vmatrix} -\lambda & 0 & 3 \\ 0 & -2-\lambda & 0 \\ -3 & 0 & -\lambda \end{vmatrix} = (-2-\lambda)\lambda^2 + 9(-2-\lambda) = -(\lambda+2)(\lambda^2+9) = 0 \ \Rightarrow \ \lambda = -2 \text{ or } \pm 3i$$

Eigenvectors corresponding to $\lambda_1 = -2$:

$$\begin{bmatrix} 2 & 0 & 3 \\ 0 & 0 & 0 \\ -3 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 2x + 3z = 0 \\ 0 = 0 \\ -3x + 2z = 0 \end{cases}$$

Since y is not part of this system, it can take any value. But the first and third equations easily imply $x = z = 0$ (solve them as you will), so that we only have the degree of freedom for y and we can choose

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Eigenvectors corresponding to $\lambda_2 = 3i$:

$$\begin{bmatrix} -3i & 0 & 3 \\ 0 & -2-3i & 0 \\ -3 & 0 & -3i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -3ix + 3z = 0 \\ (-2-3i)y = 0 \\ -3x - 3iz = 0 \end{cases}$$

The second equation implies $y = 0$ (divide by $-2-3i$ on both sides). The first and third are multiples of each other (by a factor of i, since $i^2 = -1$), providing the redundancy that this system must necessarily have. If we work with the first, we get a relation $z = ix$ and we can then take

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$$

Since our matrix is real, we can obtain an eigenvector for $\lambda_3 = -3i$ by conjugating v_2 (that is, by changing the signs of all i's in it):

$$v_3 = \begin{bmatrix} 1 \\ 0 \\ -i \end{bmatrix}$$

Example (3×3 matrix, one eigenvalue, 2-dim eigenspace). Consider

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Finding the eigenvalues:

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)^3 = 0 \;\Rightarrow\; \lambda = 2$$

Finding eigenvectors:

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad\Rightarrow\quad \left\{ \begin{array}{rcl} y+z & = & 0 \\ 0 & = & 0 \\ 0 & = & 0 \end{array} \right.$$

This system only has 1 nontrivial equation. With a 3×3 matrix, that means we'll have 2 degrees of freedom. Indeed, x and y could be anything, while z is related to y by the first equation: $z = -y$. We just need to find two independent vectors satisfying these conditions in order to get a complete system of eigenvectors. We could pick:

$$v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad , \quad w = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Example (3×3 matrix, one eigenvalue, 1-dim eigenspace). Consider

$$A = \begin{bmatrix} -1 & 4 & -1 \\ -1 & -5 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

Finding the eigenvalues:

$$\begin{vmatrix} -1-\lambda & 4 & -1 \\ -1 & -5-\lambda & 1 \\ 0 & 0 & -3-\lambda \end{vmatrix} = (\lambda+1)(\lambda-5)(-3-\lambda) + 4(-3-\lambda) = -(\lambda+3)(\lambda^2+6\lambda+5+4)$$

$$= -(\lambda+3)(\lambda^2+6\lambda+9) = -(\lambda+3)^3 = 0 \Rightarrow\; \lambda = -3$$

Finding eigenvectors:

$$\begin{bmatrix} -1+3 & 4 & -1 \\ -1 & -5+3 & 1 \\ 0 & 0 & -3+3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad\Rightarrow\quad \left\{ \begin{array}{rcl} 2x+4y-z & = & 0 \\ -x-2y+z & = & 0 \\ 0 & = & 0 \end{array} \right.$$

This system is already redundant because of the trivial last equation. There's no reason to need also that the first and second be multiples of each other, and indeed they are not. This means we'll be able to extract 2 of the variables in terms of the third and be left with only 1 degree of freedom. We can write for example

$$\begin{array}{rcl} z & = & 2x+4y \\ x & = & -2y+z = 2x+2y \;\Rightarrow\; x = -2y \end{array}$$

Choosing $y = 1$ will yield $x = -2$ and $z = 0$:

$$v = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$